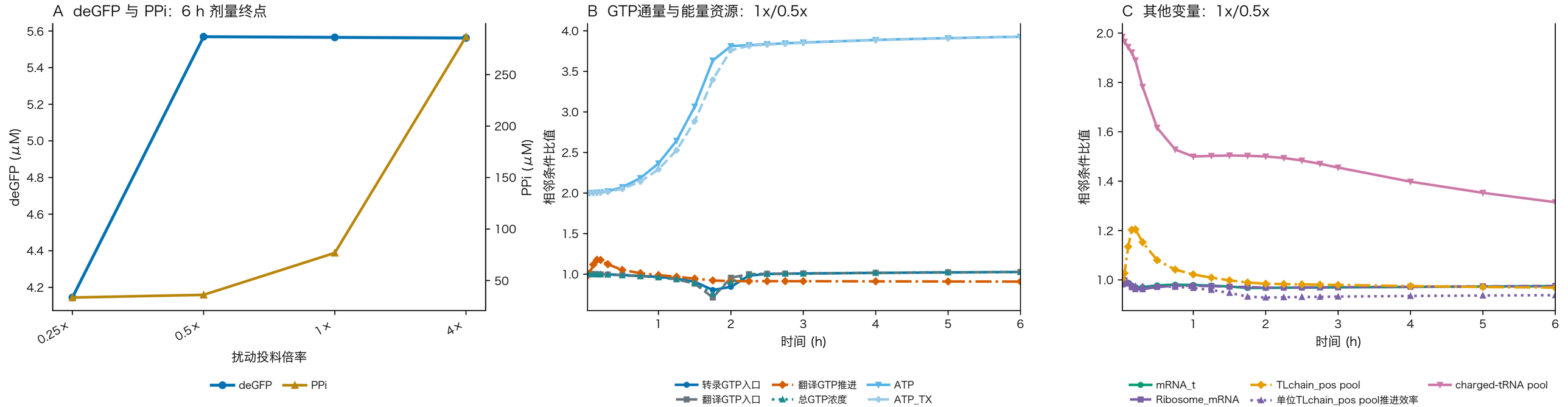ATP 单独扰动：三类曲线统一归组